



DPUATBTRFSensor

The **DPUATBTRFSensor** observes the surrounding vehicles relative to the **EGO vehicle**. The front, front left, front right as well as the rear, rear left and rear right vehicle are observed. The observation is valid in courses as well as in areas.

1. Inputs:

<i>Activate</i>	Activates the DPU (Default value: false)
<i>ScanMinDist</i>	Minimum scanbox size to the front in meters for detecting surrounding vehicles (Default: 60).
<i>v</i>	The size of the scan box is defined as maximum of ScanMinDist and the time distance calculate by a distance (see parameter THWScan) and a speed <i>v</i> in m/s (0.0).
<i>THWScan</i>	The time headway for defining the scanbox size to the front in seconds (Default: 3.0).

2. Outputs:

<i>ActiveFront/Left/Right</i>	If a vehicle was detected at the position the value is set to 1.0 otherwise to 0.0.
<i>ActiveRear/Left/Right</i>	If a vehicle was detected at the position the value is set to 1.0 otherwise to 0.0.
<i>dFront/Left/Right</i>	Distance of the front vehicle to the EGO vehicle (bumper to bumper)
<i>dRear/Left/Right</i>	Distance of the rear vehicle to the EGO vehicle (bumper to bumper)
<i>vFront/Left/Right</i>	Speed of the front/left/right vehicle in m/s.
<i>vRear/Left/Right</i>	Speed of the rear/left/right vehicle in m/s.
<i>THWFront/Left/Right</i>	Time headway between EGO vehicle and front/left/right vehicle. If the thw is undefined the value -1 is returned.
<i>TTCFront/Left/Right</i>	Time to collision between EGO vehicle and front/left/right vehicle. If the ttc is undefined the value -1 is returned.
<i>THWRear/Left/Right</i>	Time headway between EGO vehicle and rear/left/right vehicle. If the thw is undefined the value -1 is returned.
<i>TTCRear/Left/Right</i>	Time to collision between EGO vehicle and rear/left/right vehicle. If the ttc is undefined the value -1 is returned.
<i>IDFront/Left/Right</i>	Traffic ID of the front/left/right vehicle.
<i>IDRear/Left/Right</i>	Traffic ID of the rear/left/right vehicle.

3. Dependencies on other DPUs:

DPUATBTRFSensor must run on a computer together with **DPUTRF(X)**. The DPU Index entry for **DPUATBTRFSensor** must be higher than the Index entry for **DPUTRF(X)**.



4. Operations:

DPUATBTRFSensor delivers data on surrounding vehicles of the EGO vehicle. The range for detection can be set by the above parameters.

Note: In order to collect data in Area2 segments the lane neighborhoods have to be set up in the area (see reference of SILABAEEdit).

Note: Only vehicles driving on lanes in Onwards direction are scanned. Vehicles driving on oncoming lanes (in Towards direction) are not scanned. If the EGO vehicle is currently using a Towards lane itself (e.g. overtaking on a country road), all outputs are empty.

5. Example:

The DPU works well with the following script:

```
DPUATBTRFSensor Sensor
{
    Computer = {TRF};
    Index = 70;

    Activate = true;
    ScanMinDist = 100;
    THWScan = 4;
};

Connections = {
    VDyn.v -> Sensor.v
};
```